

EDGAR MEDINA

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PROFESSIONAL EXPERIENCE

GeoCam El Segundo, CA

Geocam is a multi-modal reality capture mapping system and platform. Geocam creates hardware and software that enables reality capture in indoor and outdoor environments. This technology is leveraged by local governments, telecom companies, and first responders to create digital twins of the built environment.

GIS Solutions Engineer 2022-Present

Created end-to-end custom surveying and spatial data solutions for clients by leveraging geographic, hardware, and programming knowledge in support of infrastructure, utilities, and construction environments

- Manage and maintain enterprise spatial and non-spatial databases, ensuring data quality and consistency
- Design and implement automated data workflows using Python, ArcPy, pandas, ArcGIS Python API, and GDAL for ETL processing
- Develop custom Python scripts for data validation, format conversion, and batch processing workflows
- Support telecom utility network implementations including fiber and telecommunications infrastructure mapping
- Provide technical support, troubleshooting assistance, and training for internal teams and clients with varying levels of technical experience
- Create comprehensive technical documentation including client-facing user manuals and workflow protocols
- Collaborate with project teams to coordinate surveying schedules, deliverables, and stakeholder requirements
- Develop mobile and web-based GIS applications using ArcGIS Online and custom mapping solutions
- Represent company at industry conferences to network and develop new business opportunities

IndoorVu Manhattan Beach, CA

IndoorVu collects spatial and environmental information about buildings. From facilities management, public safety, to wireless information, IndoorVu has workflows for collecting and ingesting many data formats and delivering data in a simple web application

GIS Analyst 2018-2022

Surveyed indoor environments and created 3D digital as-builts of large buildings.

- Conducted field work to investigate and validate asset locations and conditions using Trimble GNSS equipment, PCTEL radio scanners, FARO LiDAR units, and LadyBug cameras
- Processed and manipulated LiDAR point cloud data using CloudCompare and Autodesk Recap

- Produced static and interactive maps, dashboards, and visualizations using ArcMap, ArcGIS Pro, and ArcGIS Online
- Developed workflows for interoperability and data integration between CAD, BIM, and GIS platforms including Revit, AutoCAD, and ArcGIS Pro
- Published and managed web maps and feature services for client-facing applications

GIS Technician 2016-2018

Digitized floorplans using LiDAR data outputs and prepared custom web maps and dashboards for clients to visualize spatial information about buildings and surrounding environments

- Leveraged QGIS and ArcMap to process LiDAR data into raster outputs for 2D floorplan creation
- Supported spatial data design elements including schema development and cartographic standards
- Developed web-based GIS resources to enhance client communication and operational decision-making

CAL FIRE Los Osos, CA

CAL FIRE is a state agency responsible for protecting natural resources from fire. CAL FIRE works year-round to promote healthy forests and protect communities by removing overgrown vegetation.

GIS and Forestry Intern 2015-2016

Responsible for sampling flora throughout San Luis Obispo County in California to measure vegetation heat stress. Collaborated on climate research by surveying the health of Monterey Pine Trees in Cambria, CA. Produced suppression repair maps to assess post-wildfire damage.

- Used Trimble GNSS receivers to survey tree locations in forested environments
- Quantified tree mortality and tree new growth rates at Fiscalini Natural Preserve
- Converted Garmin GPS tracks to ArcMap feature classes update active fire lines
- Used Fire Incident Mapping Tools via ArcMap to produce maps used for fire briefings

Cal Poly San Luis Obispo San Luis Obispo, CA

Plant Conservatory Maintenance Steward 2014-2015

- Help maintain a diverse, well-documented, and accurately labeled living plant collection supporting and enhancing teaching and research for the faculty and students of Cal Poly
- Aide identification, cataloging, mapping, and creating signage for all the trees and large shrubs on campus

EDUCATION

A.S. Computer Science, Santa Monica College, 2023

B.S. Geography, California Polytechnic State University San Luis Obispo, 2016